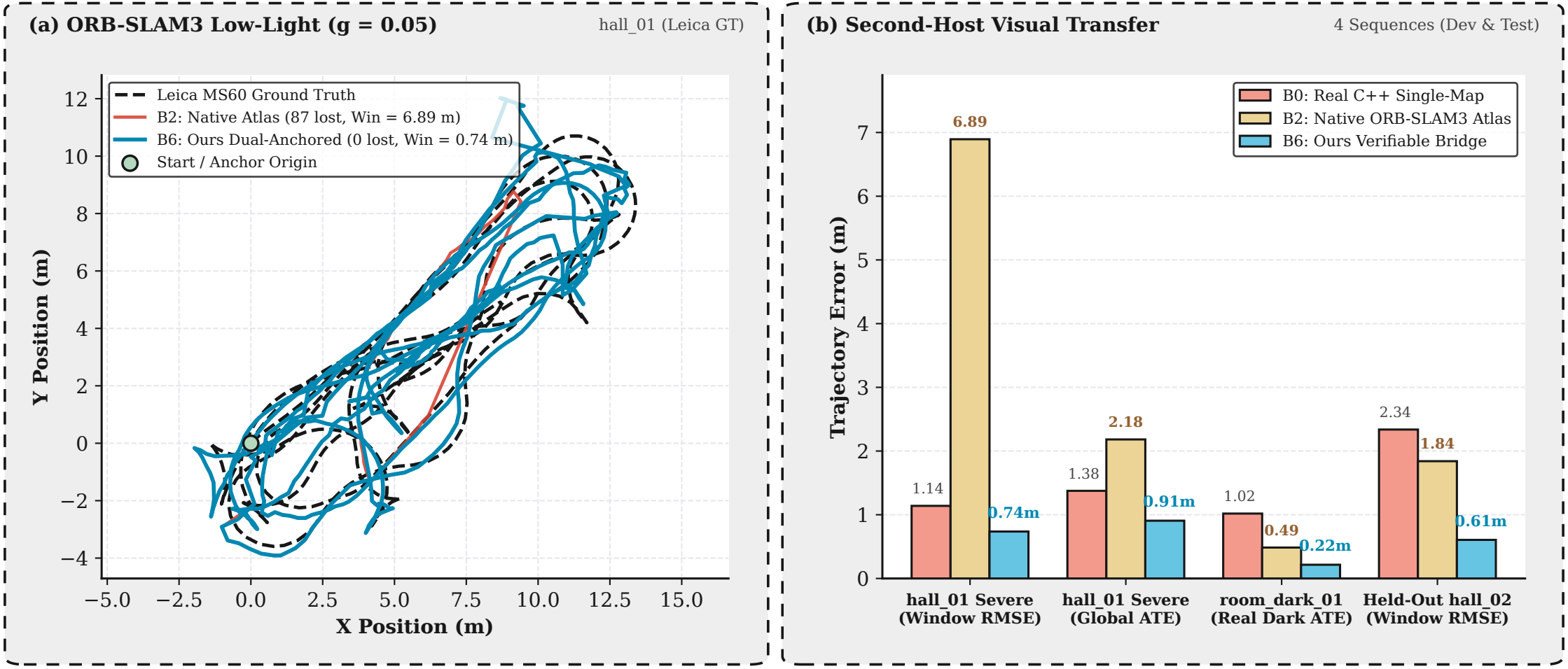

(a) ORB-SLAM3 Low-Light (g = 0.05)
hall_01 (Leica GT)
Leica MS60 Ground Truth
B2: Native Atlas (87 lost, Win = 6.89 m)
B6: Ours Dual-Anchored (0 lost, Win = 0.74 m)
Start / Anchor Origin
Y Position (m)
X Position (m)
(b) Second-Host Visual Transfer
4 Sequences (Dev & Test)
B0: Real C++ Single-Map
B2: Native ORB-SLAM3 Atlas
B6: Ours Verifiable Bridge
Trajectory Error (m)
1.14
6.89
0.74m
1.38
2.18
0.91m
1.02
0.49
0.22m
2.34
1.84
0.61m
hall_01 Severe (Window RMSE)
hall_01 Severe (Global ATE)
room_dark_01 (Real Dark ATE)
Held-Out hall_02 (Window RMSE)